Autism Detection Using Artificial Intelligence and Machine Learning

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***Abstract*– The detection of autism is a matter of immense societal and medical importance. Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition with a wide range of symptoms and severity. Early diagnosis and intervention play a pivotal role in improving the quality of life for individuals with autism. However, the diagnostic process can be time-consuming, costly, and heavily reliant on clinical expertise. With the increasing prevalence of ASD and the associated challenges in diagnosis, there is a pressing need for innovative and efficient methods to identify autism in individuals. This project aims to address this critical issue by developing an advanced autism detection system that utilizes cutting-edge technologies such as artificial intelligence and machine learning. Such a system can potentially revolutionize the early detection and intervention process, ensuring that individuals with autism receive the support they need as early as possible. Moreover, the societal and economic impact of autism cannot be understated. Autism is a lifelong condition, and its effective management and support can significantly reduce the lifetime costs for individuals and families. The ability to detect autism accurately and efficiently can lead to timely interventions, which, in turn, may improve the long-term outcomes for individuals on the autism spectrum. By harnessing the power of data analysis and AI technologies, we can enhance the accuracy and speed of autism detection, thereby easing the burden on healthcare systems and empowering caregivers and medical professionals. This project is a response to the growing global concern surrounding autism and the need for practical, data-driven solutions that can make a positive difference in the lives of individuals and families affected by autism.**

Keywords- Autism spectrum disorders, deep learning, diagnosis. Early intervention, Facial Expression Recognition.

I. INTRODUCTION

ASD affects 1.3 million children annually, with increasing rates since 1996. [1] Best practices include developmental monitoring, screening, and diagnosis. [1]

Early detection of autism spectrum disorder (ASD) is highly beneficial to the health sustainability of children. Existing detection methods depend on the assessment of experts, which are subjective and costly. In this study, we proposed a machine learning approach that fuses physiological data and behavioral data to detect children with ASD. In our autism detection project, we trained the model using 'YES/NO' questions to capture indicators of autism spectrum disorder (ASD). We introduced a diverse range of images, showcasing both individuals with autism and those without, to help the model better understand subtle facial expressions and social interactions. The model learned to recognize patterns associated with ASD traits and provide accurate assessments for early intervention. Thus, the machine learning approach proposed in this study, which combines the complementary information, can significantly improve classification accuracy.

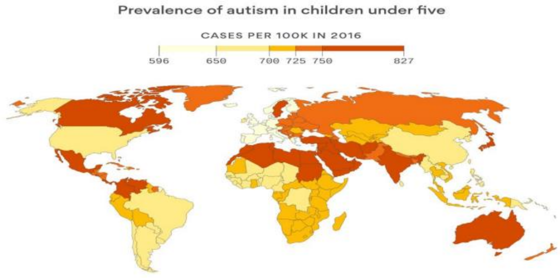
Computer vision assesses emotions, interactions, and skills in ASD children. [1] Traditional diagnostic methods have limitations in assessing ASD children. [2] Technology enhances ASD detection and assessments using ML and DL methods. [2]

Computer vision models extract joint attention skills and facial expressions in ASD children.

ASD impacts social relationships and communication skills. [1] Early intervention and medical care are crucial for ASD management. [1] ML techniques are used for early detection of ASD.

Machine learning techniques are explored for early detection and care of ASD. [1] ASD detection in early years is beneficial for reducing treatment costs.

Facial expressions are vital in communication, especially for autism children. [1].



Most of the autistic children live in country with few resources

Motivation

The motivation behind this project stems from the recognized shortcomings in current methods for diagnosing autism spectrum disorder (ASD). The prevalent issues of being time-consuming, subjective, and resource-intensive have inspired our commitment to pioneering change. Our motivation lies in the pursuit of a more efficient and objective system for early autism detection. By leveraging state-of-the-art technologies like artificial intelligence and machine learning, we aim to surmount the existing limitations and usher in a new era of accuracy and accessibility in autism diagnosis. The driving force behind our efforts is a deep-seated desire to enable timely interventions, ultimately enhancing the quality of life for individuals on the autism spectrum. This project is fueled by the belief that innovation in diagnosis can be a catalyst for transformative change, leading to more effective support and a brighter future for those affected by ASD.

Author	Algorithm	Outcome
Evelyn Fix and Joseph Hodges	K- Nearest Neighbor Algorithm	The KNN algorithm uses 'feature similarity' to predict the values of any new data points.
Joseph Berkson	Logistic Regression algorithm	The outcome must be a categorical or discrete value.
Hava Siegelman and Vladimir Vapnik	support vector clustering algorithm	If the SVC request is invalid, control is given to a special error subroutine supplied by the routine that activated the screening function.
Thomas Bayes	Naïve- Bayes	The tool combines these algorithms to evaluate autism spectrum disorder. Furthermore, the naive Bayes algorithm enhances the ASD detection process by integrating clues from different sources.

II. LITERATURE SURVEY

Reviews 33 studies, categorizes by area, support, skill development. [1] Synthesizes characteristics of distributed systems, image processing, gamification, robotics. [1] Focuses on improving social behavior, attention, communication, and reading skills. [1]
Focuses on technology's role in early detection of autism spectrum disorder. [1] Evaluates evolution, data sources, demographics, databases, controls, and outcomes. [1] Includes machine learning and deep learning for ASD risk detection. [1] Emphasizes the need for robust study protocols and multi-cultural trials.

Reviews machine learning techniques for autism spectrum disorder detection. [1] Focuses on supervised learning approaches for improved prediction accuracy. [1] Discusses the need for novel approaches to address current limitations. [1]

Various studies have been conducted that detect and diagnose ASD with different ML techniques. A rule-based Technique which accesses the ASD traits and found out that it helps in enhancing the performance. The individual features that are essential features of normal and autistic children are analyzed using tree-based classifiers. For efficient diagnoses and prognoses of ASD data by utilizing various classifiers that few features are necessary to distinguish between ASD from different attention deficit hyperactivity disorder. Most of the Autism affected children exhibit difficulty while using appropriate pragmatics [5]. The heterogeneous nature is apparent by considering the variable scores for children that receive standard language assessments. Earlier the studies have examined speech affected disorders in children on small sample analysis and recoding short speeches. Many studies have revealed that many children affected with ASD exhibit language delays which involve an extended verbal-stage delay. Of the ones who develop speech delays like echolalia which involves repeating words with no specific reason. For using automated speech recognition techniques in identifying as well as quantifying and accommodating each other's speech abnormalities with great clinical ability. The methods that are applied for assessing early ASD risk is associated with the major severity that are relevant regarding the symptoms and either helps in improvising their performance or worsen over time. Several recent researches utilize that the automated speech recordings of English-speaking children affected by Autism. Most prominent are the ones that utilize language environment analysis system (LENA) system [6].

In this section, we briefly review the related work on Autism Detection System and their different techniques. The literature survey conducted for this project revealed a rich landscape of research and advancements in the field of autism detection. Traditional diagnostic methods, while informative, have inherent limitations, including subjectivity, resource intensity, and delays in diagnosis. Recent research has witnessed a notable shift toward technology- driven solutions, prominently incorporating machine learning and artificial intelligence.

Furthermore, the utilization of various datasets, featuring a wide array of features and variables, has enabled researchers to explore innovative avenues in identifying autism with higher accuracy and efficiency.

As part of this survey, a selection of IEEE research papers were referred to, which provided valuable insights into the latest developments and methodologies in the domain of autism detection. Additionally, discussions with medical professionals, including doctors and experts specializing in autism, provided essential clinical perspectives and highlighted the need for more efficient diagnostic tools.

(a) The negative class images

(b) The positive class of eyes images

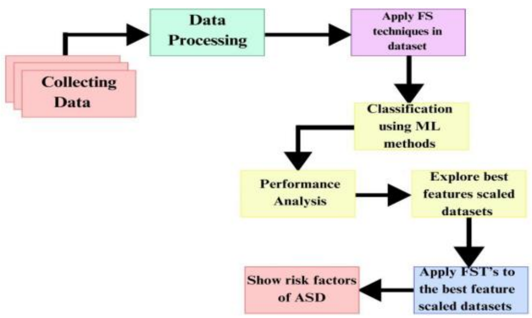
(c) The positive class of mouth images

III. METHODOLOGY

Utilizes deep learning for ASD classification using appropriate datasets. [1] Implements multimodal sensing technique for autism assessment. [2] Focuses on intervention theory and applications with ICF-based results. Feature Extraction Face, Eyes, Mouth Detection, Image Pre-processing, Facial Expression Classification. [1] Machine learning techniques used for ASD detection and classification. [1][2] Data pre-processing to enhance dataset quality and feature engineering.

Framework evaluates ML techniques for early ASD detection using various FS strategies. [1] Experiments conducted on four standard ASD datasets to compare classification outcomes. [1] Best-performing ML algorithms identified for each ASD dataset based on statistical measures. [1] Feature importance analysis guides healthcare practitioners in screening ASD cases. Implemented deep learning models for activity comprehension, joint attention, and emotions. [1] Methodology focused on diversity and fairness in data collection. [2]

Scoping review of behavioral-based ASD screening studies from 2011-2021. [1]



IV. PROPOSED METHODOLOGY

This section describes the complete proposed methodology of facial expression recognition for autistic children.

There are 4 steps in this system:

· Feature Extraction

· Facial features Detection

· Image pre-processing

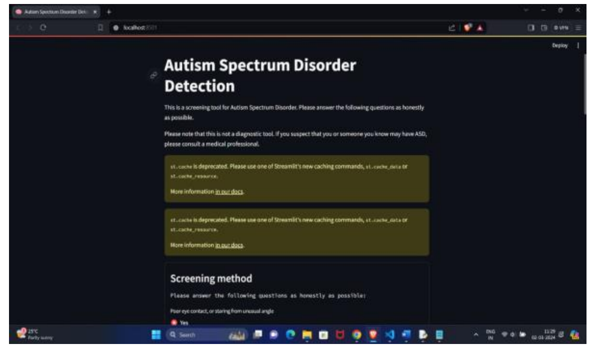
· Facial Expression Classification

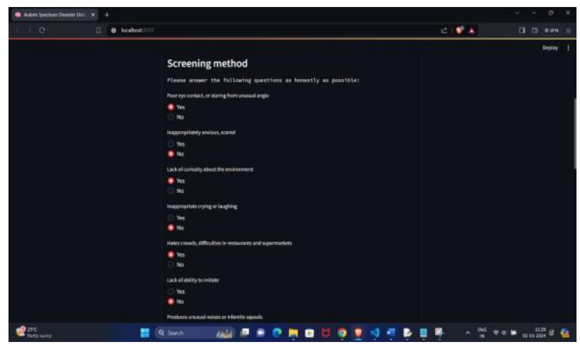
The way recognized facial expressions are through images of cropped eyes and mouths makes it a novel technique. It includes two branches: The first deals with images based in FER, and the second is input from sequential images (video) which diagnoses autism or non-autism children.

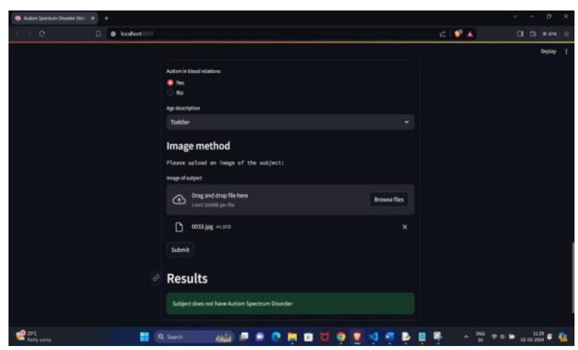
Fig. Real Dataset for Autism and Speech Problems children

V. RESULT

In the process of developing our project on autism detection, we diligently ensured that the model was trained using a collection of 'YES/NO' queries in order to capture the key indications of autism spectrum disorder (ASD). Furthermore, we integrated a range of images showcasing individuals with autism as well as individuals without autism, with the aim of enhancing the model's capability to discern subtle hints in facial expressions and social interactions. By means of iterative training, the model acquired proficiency in identifying patterns associated with ASD traits, thereby enabling it to deliver accurate assessments for early intervention.







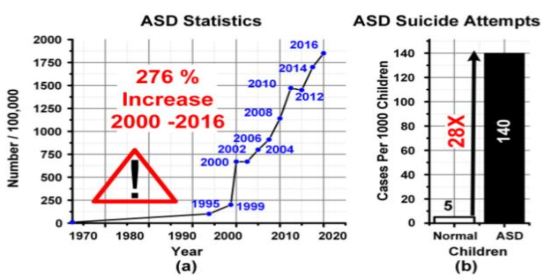
VI. CONCLUSION

In conclusion, our autism detection system represents a vital step toward a more inclusive and informed society. By harnessing the power of advanced technology, data-driven insights, and ethical considerations, we aim to enhance early diagnosis, support, and accessibility for individuals on the autism spectrum. This project not only offers a practical solution but also signifies our commitment to fostering a more compassionate and equitable world for those affected by autism. Together, we can make a positive impact and extend our hand of

understanding and support to those who need it most. thin strokes, hence, flipping pixels can be easily noticed.
Provides overview of autism diagnosis and treatment options. [1] Suggests future research directions and highlights current challenges. [1] Proposed system showed significant success with high accuracy rates. [1]

Supervised learning approaches in ASD studies show improved prediction accuracy. [1] Machine learning techniques aid in early detection and classification of ASD. [2] [1] Future studies can benefit from the research on machine learning for ASD. Feature importance analysis guides healthcare practitioners in screening ASD cases. [1] Proposed framework shows promising results for early ASD detection. [1] Models achieved high accuracy in emotion, activity, and joint attention recognition. [1] Extracted behaviours and social skills from videos for ASD assessment. [1]

Technology can enhance ASD detection process with robust study protocols. [1] Multi-cultural field trials and standardized datasets can fast-track technology adoption.



Source: A 10.13J Classification 2 Channel Deep Neural Network Based SoC for Negative Emotion Outburst Detection of Autistic Children (20 September 2021)

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